

Introductory Statistics

2025 Lectures Part 4 - Sample Space

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Experiments

- **goal:** to define **probability**, a quantitative measure of uncertainty of outcomes to occur, or the degree of belief that some proposition/conjecture is true
- rational decision maker prefers to avoid statements “I do not know what will happen” or “I do not know whether the proposition is true or false”
- probability is a tool for distinguishing likely from unlikely states of affairs
- **collected data** ... past experience, observations, results of controlled process, etc.
- data are used to build a probabilistic model of the situation and to hypothesize about mechanisms of interest

Definition 1: **Experiment** is any process, possibly under partial control, that we may observe and for which the final state of affairs cannot be specified in advance, but for which a set containing all potential final states of affairs can be identified.



Outcomes and sample space

- analyzing an experiment we observe its **outcome** – a concept which has to be specified in every situation

Definition 2: **Outcome** is a final result, observation, or measurement occurring from an experiment.

- outcomes need to
 - exclude each other
 - exhaust all logical possibilities

Definition 3: The **sample space**, denoted by S , is the set of all outcomes of an experiment. The elements of the sample space are called elementary outcomes, or sample points.

Classification of sample spaces

- finite sample space (discrete sample space)
- infinitely countable sample space (discrete sample space)
- uncountable sample space (continuous sample space)



Example 9: Consider an experiment consisting of two tosses of a regular die. Suppose that the only available information about the numbers, those that turn up on the upper faces of the die, is their sum. What is the sample space?

- outcomes can be described in various ways...the same experiment can be described via **different sample spaces**.

Example 10: Two persons enter a cafeteria and sit at a square table, with one chair on each of its sides. Suppose we are interested in the event “they sit at a corner” (as opposed to sitting across from one another). Construct a sample space.



Definition 4: An **event** is a subset of the sample space S .

Elementary event is an event that is a singleton set.

Example 9 (cont.): An event such as “the sum equals 7” containing six outcomes $(1, 6)$, $(2, 5)$, $(3, 4)$, $(4, 3)$, $(5, 2)$ and $(6, 1)$ is a subset of the sample space S .

Terminology:

- We observe the outcome. If it belongs to a set representing the event A then we say that **the event A has occurred**.
- To denote events we use letter $A, B, C...$ or $A_1, A_2...$ or as $\{X = 1\}$ or $\{a < Z < b\}$, where X and Z are some functions on sample space S



Basic operations and terminology

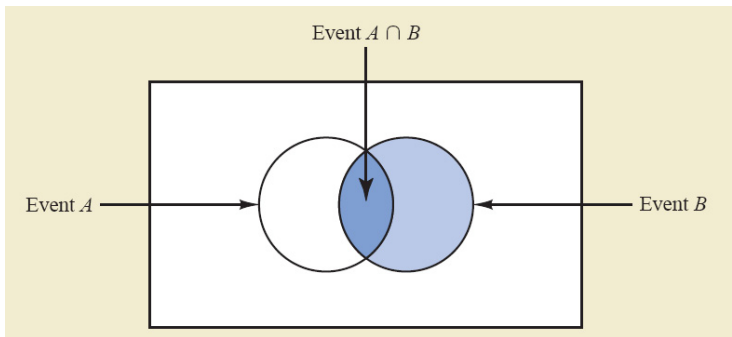
Definition 5: For events $A, B, C \subset S$ we say that

- a) $A = \emptyset$... A is **null** or **impossible event**
- b) $A = S$... A is **sure event**
- c) $A \subset B$... A is a **subevent** of B (A is contained in B or A implies B)
- d) $B = A^c = S \setminus A$... B is the **complement** of A in S
- e) $C = A \cup B$... C is the union of A and B (called event A or event B)
- f) $C = A \cap B (= AB)$... C is the intersection of A and B (called event A and event B)
- g) $C = A - B$... C occurs when A occurs and event B does not
- h) $\emptyset = A \cap B$... event A and B are **disjoint/mutually exclusive**.



Venn diagrams

- **Venn diagrams** can be used to check the validity of formulas. Picture does not constitute a proof, it may at least provide convincing evidence or suggest a method of proving the statement.
- in Venn diagram the sample space is usually represented by a rectangle and its subsets represent events.



Basic set operations

Example 11: Suppose that n shots are fired at a target, and let A_i , $i = 1, 2, \dots, n$, denote the event “the target is hit on the i th shot”. Then the union $A_1 \cup \dots \cup A_n$ is the event “the target is hit” (at least once). Its complement $(A_1 \cup \dots \cup A_n)^c$ is the event “the target is missed” (on every shot), which is the same as the intersection $A_1^c \cap \dots \cap A_n^c$.

Definition 6: (laws of operations on events)

a) Idempotence:

$$A \cup A = A, A \cap A = A$$

b) Double Complementation:

$$(A^c)^c = A$$

c) Absorption:

$$A \cup B = B \Leftrightarrow A \cap B = A \Leftrightarrow A \subset B$$



Basic set operations

d) Commutativity:

$$A \cup B = B \cup A, A \cap B = B \cap A$$

e) Associativity:

$$A \cup (B \cup C) = (A \cup B) \cup C$$

$$A \cap (B \cap C) = (A \cap B) \cap C$$

f) Distributivity:

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

g) De Morgan's Laws:

$$(A_1 \cup \dots \cup A_n)^c = A_1^c \cap \dots \cap A_n^c$$

$$(A_1 \cap \dots \cap A_n)^c = A_1^c \cup \dots \cup A_n^c$$



Example 12: (during Control Session 2)

Answer true or false.

- (i) If A and B are distinct events (i.e., $A \neq B$) such that A and B^c are disjoint, then A^c and B are also disjoint.
- (ii) If A and B are disjoint, and also B and C are disjoint, then A and C are disjoint.
- (iii) If $A \cup B^c = B^c$, then $B \subset A^c$.
- (iv) If A and B are both contained in C , then $C^c \subset A^c \cap B^c$.



Infinite sequence of events

Definitions of union and intersection can readily be extended to the case of infinite number of events:

- Let $A_1, A_2, \dots$ be an infinite sequence of events. Then

$$\cup_{i=1}^{\infty} A_i = A_1 \cup (A_2 \cup (\dots))$$

and

$$\cap_{i=1}^{\infty} A_i = A_1 \cap (A_2 \cap (\dots))$$

are events “at least one A_i occurs” and “all A_i ’s occur.”

- De Morgan’s Laws can also be extended to infinite sequences of events
- If at least one event occurs then there is one that occurs first. We can rewrite

$$\cup_{i=1}^{\infty} A_i = A_1 \cup (A_1^c \cap A_2) \cup (A_1^c \cap A_2^c \cap A_3) \cup \dots$$

as a union of disjoint events $B_i = A_1^c \cap \dots \cap A_{i-1}^c \cap A_i$ reading “ A_i is the first event in the sequence to occur”



Infinite sequence of events

Definition 7: Let $A_1, A_2, \dots$ be an infinite sequence of events. Then

$$\limsup_{n \rightarrow \infty} A_n = \bigcap_{k=1}^{\infty} \bigcup_{i=k}^{\infty} A_i, \quad \liminf_{n \rightarrow \infty} A_n = \bigcup_{k=1}^{\infty} \bigcap_{i=k}^{\infty} A_i$$

are the events that “infinitely many A_i ’s occur” and that “all except finitely many A_i ’s occur”, respectively.

- Clearly, if all but finitely many event occur then infinitely many events occur, i.e.

$$\limsup_{n \rightarrow \infty} A_n \supset \liminf_{n \rightarrow \infty} A_n$$

Definition 8: If $\limsup_{n \rightarrow \infty} A_n \subset \liminf_{n \rightarrow \infty} A_n$ then we say that the sequence $\{A_n\}_{n=1}^{\infty}$ converges and the **limit of events** is given as

$$\lim_{n \rightarrow \infty} A_n = \limsup_{n \rightarrow \infty} A_n = \liminf_{n \rightarrow \infty} A_n$$



Infinite sequence of events

Theorem 1:

- a) If the sequence of A_i 's is increasing ($A_1 \subset A_2 \subset \dots$) then the limit $\lim_{n \rightarrow \infty} A_n$ exists, i.e.

$$\liminf_{n \rightarrow \infty} A_n = \limsup_{n \rightarrow \infty} A_n = \lim_{n \rightarrow \infty} A_n, \text{ and}$$

$$\lim_{n \rightarrow \infty} A_n = \bigcup_{n=1}^{\infty} A_n.$$

- b) If the sequence of A_i 's is decreasing ($A_1 \supset A_2 \supset \dots$) then the limit $\lim_{n \rightarrow \infty} A_n$ exists, i.e.

$$\liminf_{n \rightarrow \infty} A_n = \limsup_{n \rightarrow \infty} A_n = \lim_{n \rightarrow \infty} A_n, \text{ and}$$

$$\lim_{n \rightarrow \infty} A_n = \bigcap_{n=1}^{\infty} A_n.$$

Example 13: Find $\lim_{n \rightarrow \infty} A_n$ if

- (i) $A_n = \{x | \frac{1}{n} \leq x \leq 3 - \frac{1}{n}\}, n = 1, 2, 3, \dots$
- (ii) $A_n = \{x | 2 < x \leq 2 + \frac{1}{n}\}, n = 1, 2, 3, \dots$
- (iii) $A_n = \{(x, y) | 0 \leq x^2 + y^2 \leq \frac{1}{n}\}, n = 1, 2, 3, \dots$



Need for σ -algebras

- in order to define properly probability, we need to understand numerical functions on subsets of sample space
 - in case of finite or countable S we can consider the class of all subsets (and it will not cause any troubles for definition of probability)
 - the system of all subsets of uncountable S is too rich, it contains “special” sets with nonunique “length” and thus probability defined on all subsets may produce paradoxes – nonunique values of probability of some subsets called **nonevents** – we need restrictions to a system of subsets which does not contain such subsets and is closed under basic operations of unions, intersections and complements

Definition 9: (closure under operation)

We say that a class $\mathcal{A}$ of subsets of S is closed under a given operation if the sets resulting from performing this operation on elements of $\mathcal{A}$ are also elements of $\mathcal{A}$.



Definition 10: Let S be a nonempty set. Then nonempty class $\mathcal{A}$ of subsets of S is called σ -algebra (σ -field), if

- (i) $A \in \mathcal{A} \Rightarrow A^c \in \mathcal{A}$ (closed on complements)
 - (ii) $A_n \in \mathcal{A}, n = 1, 2, \dots \Rightarrow \bigcup_{n=1}^{\infty} A_n \in \mathcal{A}$ (closed on countable unions)
- Closeness on finite/countable intersections follows from de Morgan's laws.

Example 14: Let $S_1 = \{a\}$ and $S_2 = \{a, b\}$. Construct σ -algebras of all subsets.

Example 15: Let $S = \{1, 2, \dots\}$ and let $\mathcal{A}$ consists of all subsets of S that are finite. Is $\mathcal{A}$ a σ -algebra?

Example 16: Let $S = \{1, 2, \dots\}$, and let $\mathcal{A}$ be the class of all subsets A of S such that either A or A^c is finite. Is $\mathcal{A}$ a σ -algebra?



Definition 11: Let S be any set.

- (i) The class consisting of two sets, $\emptyset$ and S , is called the **smallest** σ -algebra.
- (ii) The class of all subsets of S is called the **largest** (or maximal) σ -algebra.
- For any event A it is easy to check that the class $\mathcal{A}$, consisting of the four events $\emptyset, A, A^c, S$, is closed under any operations. This class is an example of a σ -algebra that contains the events A and A^c , and it is the smallest such σ -algebra.



The following theorem ensures existence of σ -algebra to any collection of subsets.

Theorem 2: (without proof) For any nonempty class $\mathcal{K}$ of subsets of S there exists $\sigma(\mathcal{K})$, a unique smallest σ -algebra containing all sets in $\mathcal{K}$. It is called the σ -algebra generated by $\mathcal{K}$.

Why not to take the maximal σ -algebra in the case of uncountable S ?

On a real line it is natural to consider “simple” events in the form of intervals. σ -algebra generated by all open intervals is called **Borel** σ -algebra $\mathcal{B}(\mathbb{R})$. It is of a rich structure. On the other hand, the maximal σ -algebra on $\mathbb{R}$ is too rich and contains “nonevents” - e.g. Vitali sets.

